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# A full-scale Monte Carlo Potts model and real time conversion under non-uniform temperature distribution

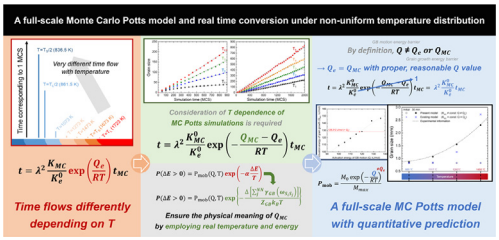
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## HIGHLIGHTS

- Monte Carlo Potts model is modified to fully consider the real temperature and energy values.
- Real time conversion scheme for Monte Carlo Potts model is modified to be temperature-independent.
- The present model can avoid the time conversion error, and furthermore, reproduces grain growth more accurately than the existing model.
- Monte Carlo Potts model now can predict grain growth in various materials under various temperature condition in a quantitative manner.

## GRAPHICAL ABSTRACT




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## ABSTRACT

Prediction of microstructure evolution is important to design structural materials, and grain growth is the most fundamental but important one which determines mechanical properties. The Monte Carlo Potts is a widespread simulation model for grain growth with high computational efficiency but has not been considered suitable for the quantitative prediction of practical processes since the physical meaning of variables is not fully understood yet. Efforts have been made to assign realistic physical meaning to variables. The existing equation for converting simulation time to real time involves a strong temperature dependence, which causes a problem under non-uniform temperature distribution, as in additive manufacturing. This problem could be solved by assigning a temperature dependence to the simulation result, which was possible by using realistic temperature and energy values. We found that the simulated activation energy for grain growth could be equated to the experimental value by adjusting the activation energy for grain boundary migration within a reasonable range in the simulation. The modified model made the time conversion equation completely temperature-independent and was free from the above-mentioned problem in time conversion. The Monte Carlo Potts simulation can now more reasonably predict grain growth in various materials under various temperature conditions.

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## 1. Introduction

Predicting and controlling the evolution of microstructure is important to design optimal structural materials with robust mechanical properties [1–3], and grain growth is the most fundamental but important microstructure evolution which significantly affects the mechanical properties of polycrystalline materials [4–6]. Recently, it has become even more important with increasing interest in the metal additive manufacturing (AM) technique, which produces different microstructures from conventional processes [7–9]. The laws of grain growth kinetics in two-

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dimensional (2D) space (such as the parabolic law [10] and the von Neumann-Mullin law [11,12]) have been deeply understood and also have been expanded into microstructure evolution in 3D space [13,14]. Because it is not simple work to understand the complex nature of grain boundary (GB) networks and the interaction between the grains in 3D [15–17], computer simulation models (such as Monte Carlo (MC) Potts model [18], phase field model [19–21], surface evolver model [22], and vertex model [23]) have been widely used to provide insights into the details of grain growth [24–27]. Among them, the MC Potts is a widely used model that enables simulations of various microscopic phenomena using a simple algorithm and low computational cost [28,29]. It is also known that MC Potts simulations agree well with analytical solutions for grain growth such as parabolic growth law [30] and the 3D von Neumann equation [25,26].

Although the MC Potts model has greatly contributed to understanding microstructure evolution, it is not considered suitable for quantitative prediction and design of practical processes since the physical meaning of variables (time, length, temperature, and energy) is not fully understood yet, and the model employs "unitless" simulation variables. Several studies have been done to link the simulation variables to the real world. Gao and Thompson [31] proposed a real time–temperature model which can assign real length and time scales to the MC Potts space, and Raabe [32] proposed introducing a switching function into the transition function to further consider the temperature dependence of grain growth in the simulation.

Despite such efforts, the MC Potts model still cannot directly deal with real energy and temperature since the physical scale of GB energy is not yet clear. That is, the model cannot fully evaluate the contribution of the driving force of grain growth (minimization of GB energy) in simulating grain growth. Moreover, this critical point also ruins the real time conversion scheme [31] under a non-uniform temperature distribution condition, making the MC Potts model unsuitable for quantitative prediction under practical process conditions. For example (which will also be discussed in detail later on), Rodgers *et al.* [28] conducted an MC Potts study to quantitatively predict microstructure evolution during the metal AM process by adopting the real time–temperature model [31], but they could not consider the temperature dependence of grain growth in the simulation space into the real time–temperature model due to above-mentioned physical ambiguity of the MC Potts model. This led to an unrealistic situation that the time length corresponding to one MC step (MCS) becomes significantly different even within an AM sample, as shown in Fig. 1. Thus, the MC Potts model should be further complemented to predict and design such a practical process in a realistic, quantitative manner.

In the present work, we propose an MC Potts model that can fully consider the grain growth kinetics by dealing with realistic GB energy information and real temperature. Then, we propose a scheme for temperature-independent real time conversion by considering the temperature dependence of simulation results. Furthermore, we confirm that the modified real time conversion scheme reproduces grain growth much more accurately than the existing model.

## 2. Original MC Potts model and realistic scale assignment

### 2.1. Original MC Potts model

The MC Potts model consists of random attempts and their acceptance according to transition probability. Attempts equal to the number of grid points in a simulation sample are referred to as one Monte Carlo Step (MCS), a time unit in the MC Potts simulation. The transition probability is usually calculated by the Metropolis function:

$$P_{df}(E,T) = \begin{cases} \exp\left(-\alpha\frac{\Delta E}{T}\right) & if\,\Delta E > 0 \\ 1 & if\,\Delta E \le 0 \end{cases} \quad (1)$$

where $\alpha$ is an arbitrary constant, $\Delta E$ is the change in total energy of the system caused by an attempt, and $T$ is the temperature. This function considers the driving force of grain growth where the driving force of grain growth is the minimization of grain boundary (GB) energy. The transition probability in Eq. (1) is always calculated to be 100 % when the attempt is energetically favorable. This leads to unrealistic simulation results such as grain growth at 0 K, which indicates that the temperature dependence of the grain growth related to GB motion by atomic diffusion is not fully considered in the model. Raabe [32] proposed introducing a temperature-dependent function into the transition probability, as in Eq. (2), and Rodgers *et al.* [33] further developed it with the GB mobility ($M$) as in Eq. (3).

$$P = P_{mob}(M)P_{df}(E,T) \quad (2)$$

$$P_{mob} = \frac{M_0 \exp\left(-\frac{Q}{RT}\right)}{M_{max}} = \exp\left[-\frac{Q}{R}\left(\frac{1}{T}-\frac{1}{T_m}\right)\right] \quad (3)$$

$Q$ is the activation energy of the GB migration, R is the gas constant, and $T_m$ is the maximum temperature. In this case, the probability of grain growth becomes 0 at 0 K ($\lim_{T\to 0}\exp(-Q/RT) = 0$), and the above-mentioned error is naturally corrected.

### 2.2. Real time–temperature equation for MC Potts model

Almost all MC Potts studies employ "unitless" simulation variables and have used simulation results only for qualitative comparison and analysis because the physical meaning and scale of simulation variables (time, length, temperature, and energy), especially for the driving force term ($P_{df}$) shown in Eq. (1), are not fully understood yet. Several studies have been conducted to link the simulation variables with the real world. Gao and Thompson [31] proposed a real time–temperature model with a simple MC Potts model which does not have any temperature dependence in the transition probability. Their work showed that even a simple MC Potts model reproduces the grain growth well and also showed that real length and time scales can be assigned to the MC Potts space. The model is briefly described below.

The grain growth can be described using the following equation.

$$L(t)^2 - L(0)^2 = K_e t \quad (4)$$

where $t$ is the time, $L(t)$ is the grain size as a function of time, and $K_e$ is the temperature-dependent factor for the grain growth. This equation is also valid in the MC Potts simulation.

$$L_{MC}(t_{MC})^2 - L_{MC}(0)^2 = K_{MC} t_{MC} \quad (5)$$

$t_{MC}$ is the simulation time, $L_{MC}(t_{MC})$ is the grain size in the simulation length unit, and $K_{MC}$ is the grain growth factor in the simulation. The real length and the simulation length can be linked as follows:

$$L = \lambda L_{MC} \quad (6)$$

where $\lambda$ is the real distance between grid points in the simulation. Then, the relationship between simulation time and real time can be obtained by equating Eq. (4) and Eq. (5), ignoring the $L(0)$ and $L_{MC}(0)$ terms for simplicity:

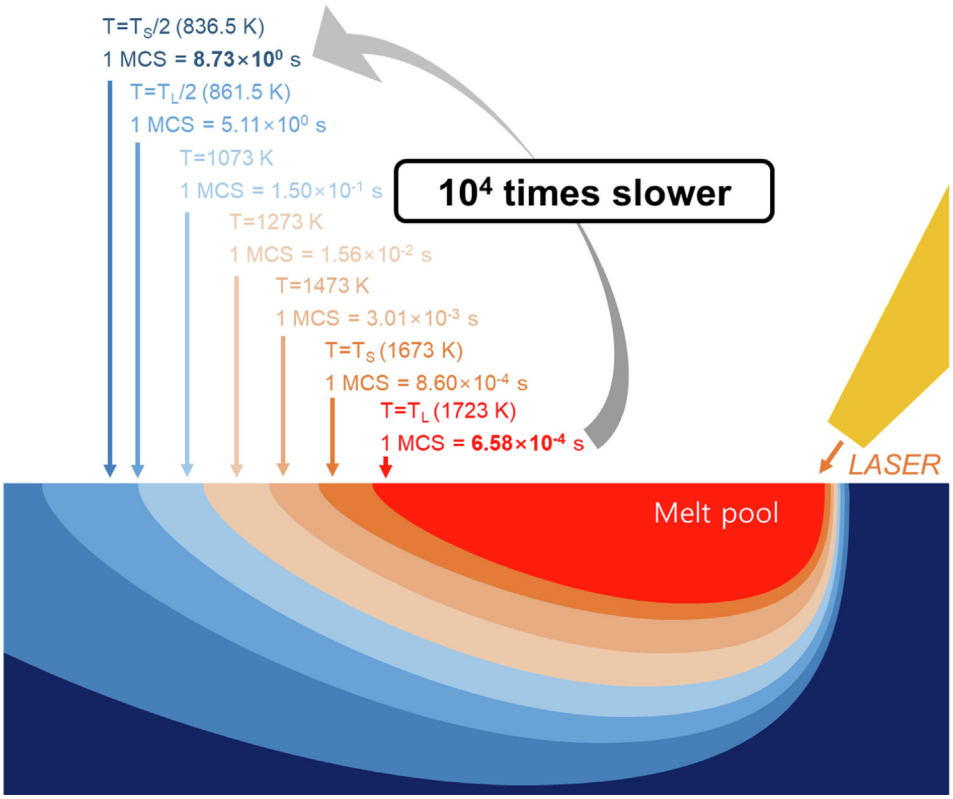



**Fig. 1.** Converted real time corresponding to one MCS at different locations within a 304L stainless steel AM sample with a qualitatively assumed temperature profile. Note that $\lambda$, $K_e^0$, $Q_e$, and $K_{MC}$ were 2.5 μm, 2.04133 × $10^{-5}$ $m^2$/s, 128,312 J/mol, and 0.27695 point$^2$/MCS, respectively, the same as used by Rodgers *et al.* [28].

$$t = \lambda^2 \frac{K_{MC}}{K_e} t_{MC} \tag{7}$$

$K_e$ can be determined from experimental data, and its temperature dependence is usually expressed using an Arrhenius-type function:

$$K_e = K_e^0 \exp\left(-\frac{Q_e}{RT}\right) \tag{8}$$

where $K_e^0$ is the pre-exponent factor, and $Q_e$ is the activation energy of the grain growth. Then, Eq. (7) changes to:

$$t = \lambda^2 \frac{K_{MC}}{K_e^0} \exp\left(\frac{Q_e}{RT}\right) t_{MC} \tag{9}$$

### 2.3. Temperature dependence of real time conversion equation

Recently, Rodgers *et al.* [28,34] employed the transition probability function in Eq. (2) for the MC Potts simulation of AM processes and applied the above model (Eq. (9)) to assign the realistic time and length scale to the simulation, reproducing grain morphology and size distribution similar to experimental results. However, there could be a difficulty in the real time conversion.

Both Gao and Thompson [31] and Rodgers *et al.* [28] treated the factor for grain growth in the simulation ($K_{MC}$) as a constant. In this case, the ratio $t/t_{MC}$ in Eq. (9) becomes temperature dependent proportional to $\exp(Q_e/RT)$, and the time scale corresponding to one MCS significantly varies within a sample where temperature varies severely. For instance, Fig. 1 shows how significantly the time corresponding to one MCS can vary in an AM sample. That is, time may flow differently within a single sample when the temperature distribution is non-uniform. This will lead to physically illogical simulation results, especially for AM samples. Some studies [35,36] have tried to avoid this problem by adjusting the number of random selections differently depending on the temperature, referring to the length of time corresponding to one MCS. Such modeling is a bit questionable as it is no longer based on "random" selections: the statistical basis inherent in MC methods may not be valid, so it may not be a proper solution.

We noticed that the most fundamental cause of the problem lies in assuming $K_{MC}$ to be a constant. According to the transition probability function of the MC Potts model (the original model in Eq. (1) or the modified model in Eq. (2)), the simulation results are expected to have a certain temperature dependence. For example, Fig. 2 shows a typical temperature dependence of grain growth

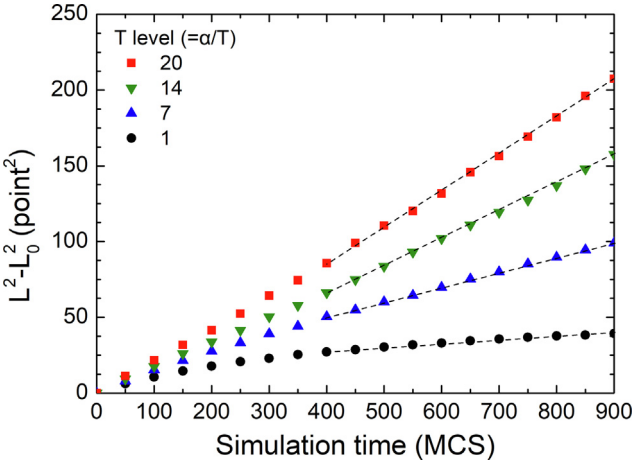



**Fig. 2.** Simulated grain size as a function of simulation time at different simulation temperatures using the original Potts model (Eq. (1)).

kinetics that can be obtained in MC simulations according to the original Potts model (Eq. (1)). That is, $K_{MC}$ should also be a function of temperature. Here, $K_e$ and $K_{MC}$ are quantities having analogy, so $K_{MC}$ can be naturally expected to have an Arrhenius function type temperature dependence like $K_e$:

$$K_{MC} = K_{MC}^0 \exp\left(-\frac{Q_{MC}}{RT}\right) \tag{10}$$

where $K_{MC}^0$ is the pre-exponent factor, and $Q_{MC}$ is the activation energy of grain growth obtained from the simulation results. Taking Eq. (10) into Eq. (9) leads to:

$$t = \lambda^2 \frac{K_{MC}^0}{K_e^0} \exp\left(-\frac{Q_{MC} - Q_e}{RT}\right) t_{MC} \tag{11}$$

In this case, the temperature dependence of the real time corresponding to one MCS can be far reduced. Here, the length and time are the quantities that need scaling and conversion but the temperature and energy are not quantities that need scaling or conversion. Then, one can even think that $Q_e$ and $Q_{MC}$ represent the same quantity and should have the same value once the MC Potts model fully describes the grain growth kinetics in the real world.

Even though the $K_{MC}$ has a temperature dependence as described above, Gao and Thompson [31] and Rodgers *et al.* [28] treated it as a constant. The present authors believe that this was because they could not assign realistic values to the GB energy and thus the temperature in Eq. (1). $K_{MC}$ represents the grain growth kinetics in the MC Potts simulation resulting from the transition probability function. If the transition probability function cannot deal with realistic temperature and energy values, the temperature dependence in the simulated grain growth ($K_{MC}$) would lose the physical significance, and the resultant activation energy of reaction ($Q_{MC}$) would be nothing but a hypothetical variable. This may be the reason why those authors had to treat $K_{MC}$ as a constant without any temperature dependence.

## 3. Development of a full-scale MC Potts model

### 3.1. Modified MC Potts model dealing with realistic variables

As stated in Section 2.3, any physical meaning in temperature dependence of simulation results could not be ensured if the transition probability function cannot deal with realistic GB energy and temperature values. Accordingly, we first assigned realistic values to the GB energy and temperature. By definition, an Arrhenius-type probability function for $N$ atoms can be expressed as follows:

$$P_{df}(\Delta E > 0, T) = \exp\left(-\frac{\Delta E}{N k_B T}\right) \tag{12}$$

where $k_B$ is the Boltzmann constant. Considering the grain growth only, the change in the total energy of the system ($\Delta E_i$) can be calculated as the change in GB energy.

$$\Delta E = \Delta\left[\sum_j^{NN} \gamma_{GB}\left(\omega_{S_iS_j}\right) A_{ij}\left(1 - \delta_{S_iS_j}\right)\right] \tag{13}$$

$A_{ij}$ is the area of GB between site $i$ and $j$, $S_i$ is the orientation of the grain on site $i$, $\omega_{S_iS_j}$ represent the misorientation and inclination angles between site $i$ and $j$, $\gamma_{GB}$ is the GB energy as a function of $\omega_{S_iS_j}$, and $\delta_{ij}$ is the Kronecker delta function. Considering that the $(1 - \delta_{S_iS_j})$ term can be included in the GB energy expression (i.e. $\gamma_{GB}(0) = 0$), taking Eq. (13) into Eq. (12) leads to:

$$P_{df}(\Delta E > 0, T) = \exp\left\{-\frac{1}{k_B T}\Delta\left[\sum_j^{NN} \frac{\gamma_{GB}\left(\omega_{S_iS_j}\right) A_{ij}}{N_{ij}}\right]\right\} \tag{14}$$

In order to deal with accurate GB energy, it is necessary to consider the misorientation and inclination angles and utilize GB databases that are organized based on such information [37,38]. However, in the present work, GB energy was just assumed to be a function of a single disorientation for simplicity. Further, we assumed that GB energy was only related to the atoms in the GB region, in which case $N_{ij}$ is the number of atoms in the GB region between site $i$ and $j$. Each site in MC Potts simulation is treated as a finite grid cell of the same shape, so the value of $A_{ij}$ is the same for all sites. $N_{ij}$ will slightly vary depending on the type of GB, but again we did not consider the type of GB at this moment and assumed that $N_{ij}$ has a constant average value for all GBs. Then, $N_{ij}/A_{ij}$ can be reduced to $Z_{GB}$, the average number of atoms per unit area of the GB region (i.e., the areal density of atoms on GB).

$$P_{df}(\Delta E > 0, T) = \exp\left\{-\frac{\Delta\left[\sum_j^{NN} \gamma_{GB}\left(\omega_{S_iS_j}\right)\right]}{Z_{GB} k_B T}\right\} \tag{15}$$

With Eq. (15), the transition probability can consider the real GB energy and temperature. Finally, the transition probability function for the grain growth can be organized as follows:

$$P = \begin{cases} \exp\left[-\frac{Q}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)\right] \exp\left\{-\frac{\Delta\left[\sum_j^{NN} \gamma_{GB}\left(\omega_{S_iS_j}\right)\right]}{Z_{GB} k_B T}\right\} & \text{if } \Delta E > 0 \\ \exp\left[-\frac{Q}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)\right] & \text{if } \Delta E \le 0 \end{cases} \tag{16}$$

Now the transition probability can fully consider both the kinetics and energetics of grain growth based on fundamental materials properties.

### 3.2. Modeling scheme for temperature-independent real time conversion equation

Next, we examined the temperature dependence of the grain growth kinetics in the simulation ($Q_{MC}$ in Eq. (10)) using the modified model with realistic simulation parameters. As an example, 304L stainless steel, whose grain growth kinetics had been experimentally investigated [39], was simulated, as also had been done by Rodgers *et al.* [28]. Four simulation parameters had to be determined: $Q$, $T_m$, $\gamma_{GB}$, and $Z_{GB}$. $Q$ was first set to be the same as the experimentally obtained activation energy of grain growth ($Q_e$, 128,312 J/mol), which was taken by Rodgers *et al.* [28]. $T_m$ was set to be the liquidus temperature (1723 K). For $\gamma_{GB}$, the average value of high-angle grain boundary energy was calculated to be 1.3–1.4 J/m$^2$ in atomistic studies for pure γ-Fe [37,40], and the average value of $\gamma_{GB}$ experimentally measured for the 304L steel was similar (or a bit larger) to that of pure γ-Fe [41]. Accordingly, $\gamma_{GB}^{max}$ was set to be 1.4 J/m$^2$, and as mentioned earlier, GB energy was assumed to be a simple function of a single disorientation at this moment. $Z_{GB}$ was calculated to be $8.96 \times 10^{19}$ atoms/m$^2$ from the atomic density of γ-Fe [42] and the assumed GB thickness (10 Å), assuming that the atomic density in the GB region of γ-Fe is not far different from that in the bulk. Other simulation details are given as Supplementary Information.

The simulation results at various temperatures are presented in Fig. 3. According to the $K_{MC}$ values obtained at each temperature, $Q_{MC}$ was calculated to be 143,370 J/mol, which is similar to but not exactly the same as the experimental value ($Q_e$, 128,312 J/mol). It should be noted here again that $Q$ is the "pure" activation energy barrier of GB migration, excluding the influence of driving force on grain growth, whereas $Q_e$ (and also $Q_{MC}$) is the activation energy barrier of overall grain growth contributed from both the GB motion and the driving force. Considering the

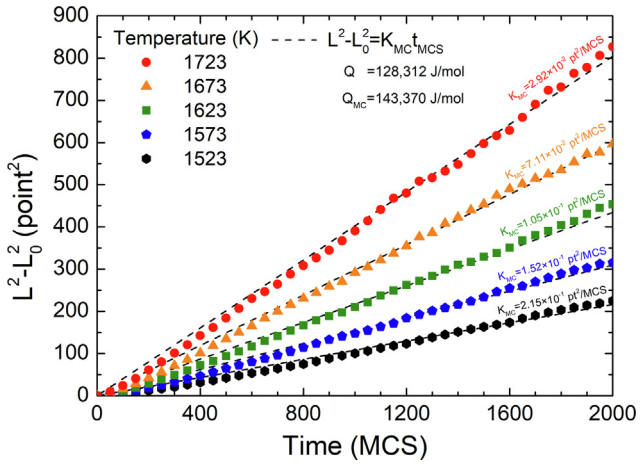



**Fig. 3.** Simulated grain growth in a 304L stainless steel at different temperatures using the modified MC Potts model (Eq. (16)). Activation energy of GB migration ($Q$) was set to be equal to the experimentally observed activation energy of grain growth ($Q_e$).

physical meaning of $Q$ and $Q_e$, we believe that $Q$ does not necessarily have to be equal to $Q_e$. The exact value of $Q$ is difficult to determine only from the grain size data, unlike $Q_e$ or $Q_{MC}$. Although we initially used the same value for $Q$ as $Q_e$ in Fig. 3, it was not a strict requirement and the $Q$ value could be adjusted so that the resultant $Q_{MC}$ value becomes the same as $Q_e$.

In order to determine the $Q$ value, a large-scale atomistic simulation [43,44] may be an efficient approach with reasonable physical basis, but at this moment we just employed a trial-and-error approach to focus on demonstrating the solution of the time conversion error. In order to see the possibility of adjusting the $Q$ value in a reasonable range, the effect of $Q$ value on the resultant $Q_{MC}$ was examined. Fig. 4 shows that $Q_{MC}$ varies proportionally to $Q$, and a $Q_{MC}$ value equal to $Q_e$ (128,312 J/mol [28]) is obtained when shifting the $Q$ value from 128,312 to 118,026 J/mol. Generally, the activation energy of GB migration, $Q$, is known to be similar to the activation energy of GB diffusion ($Q_{GB}$) for high-angle GBs [45–48], while some computational studies claim that $Q$ may be even lower than $Q_{GB}$ [49–51]. The experimentally measured value of the activation energy of $Q_{GB}$ in γ-Fe is in a range of 100.7 – 174.6 kJ/mol [52–54]. Accordingly, the determined $Q$ value, 118.0 kJ/mol, is thought to be reasonable, in accordance with common knowledge.

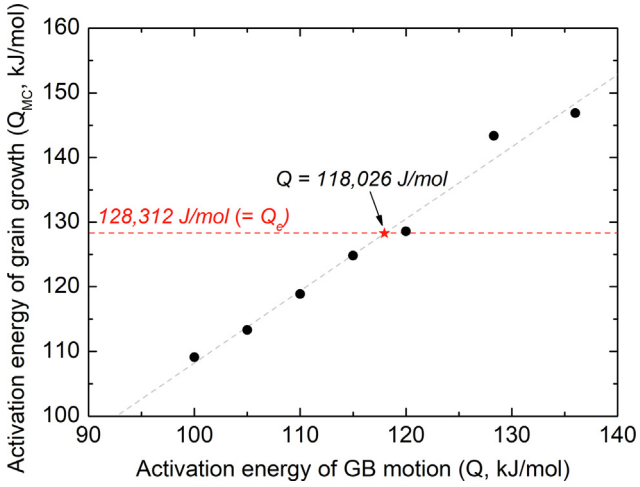



**Fig. 4.** Activation energy of grain growth ($Q_{MC}$) obtained from simulation results for 304L stainless steel with different activation energies of GB migration ($Q$) values.

With such a $Q$ value that makes $Q_{MC}$ equal to $Q_e$, the real time conversion becomes temperature-independent, as follows:

$$t = \left[\lambda^2 \frac{K_{MC}^0}{K_e^0}\right] t_{MC} \tag{17}$$

Then, the problem in the real time conversion shown in Fig. 1 is completely solved. Overall, the above procedure to avoid the time conversion problem can be summarized as follows:

1) Take all simulation variables (except for $Q$) from fundamental materials property information in literature, including information for a grain growth factor ($K_{MC}$: $K_e^0$ and $Q_e$).

2) Through MC Potts simulations with different $Q$ values, find the $Q$ value that makes the resultant activation energy in MC Potts simulation ($Q_{MC}$) equal to the experimental value ($Q_e$).

Finally, to confirm the validity of the present modeling, we simulated grain growth for a 304L stainless steel at various temperatures and compared the results with those from the existing modeling described in Section 2 (which uses a constant value for $K_{MC}$ in Eq. (9) and the value of $Q_e$ for $Q$ in Eq. (16)). As shown in Fig. 5, the present model well reproduces the experimental information [39] whereas the existing model shows a discrepancy. It should be emphasized here that such an improvement in the accuracy of quantitative prediction originates from making the time conversion equation temperature-independent (by considering the temperature dependence of $K_{MC}$). That is, although the main focus and achievement in the present work was the applicability of the MC Potts model to non-uniform temperature distribution conditions, the accuracy of quantitative prediction could also be further improved.

Consequently, the modified MC Potts model and the real time conversion scheme proposed in the present work are significant in that they not only solve the time conversion problem under non-uniform temperature distribution but also provide much more reliable results in a quantitative manner. Since all the simulation variables were determined by considering their physical meaning and obtained from fundamental materials properties, we believe that the present model is generally applicable to various polycrystalline materials. Now the simulation results are expected to show reasonable grain growth kinetics even under non-uniform temperature distribution conditions.

It was shown by Garcia *et al.* [55] that the local grain growth kinetics under non-uniform temperature field with steady-state heat flow (i.e. linear temperature profile condition) is identical to that under isothermal condition at corresponding temperature. That is, if the model well reproduce the isothermal grain growth kinetics, actual grain growth kinetics is also expected to be reasonable even under non-uniform temperature distribution. Accordingly, the present model is expected to reproduce a reasonable grain growth kinetics under non-uniform temperature distribution. Since the present time conversion equation is temperature-independent, it is now available to directly compare simulation results with practical processes, such as AM process. Such a simulation further requires other simulation techniques (e.g. solver for the heat transfer equation, solidification model). In future work, the present model will be utilized to directly predict microstructure evolution during practical AM processes. This will be a useful computational method to optimal AM process design with high computational efficiency.

On the other hand, Raabe [32] proposed that the transition probability function with the mobility term (which is also adopted in the present model: see Eq. (2)) is applicable to other types of microstructure evolution. We believe that MC Potts models for other types of microstructure evolution can also be modified to deal with realistic variables as in the present work. This will also

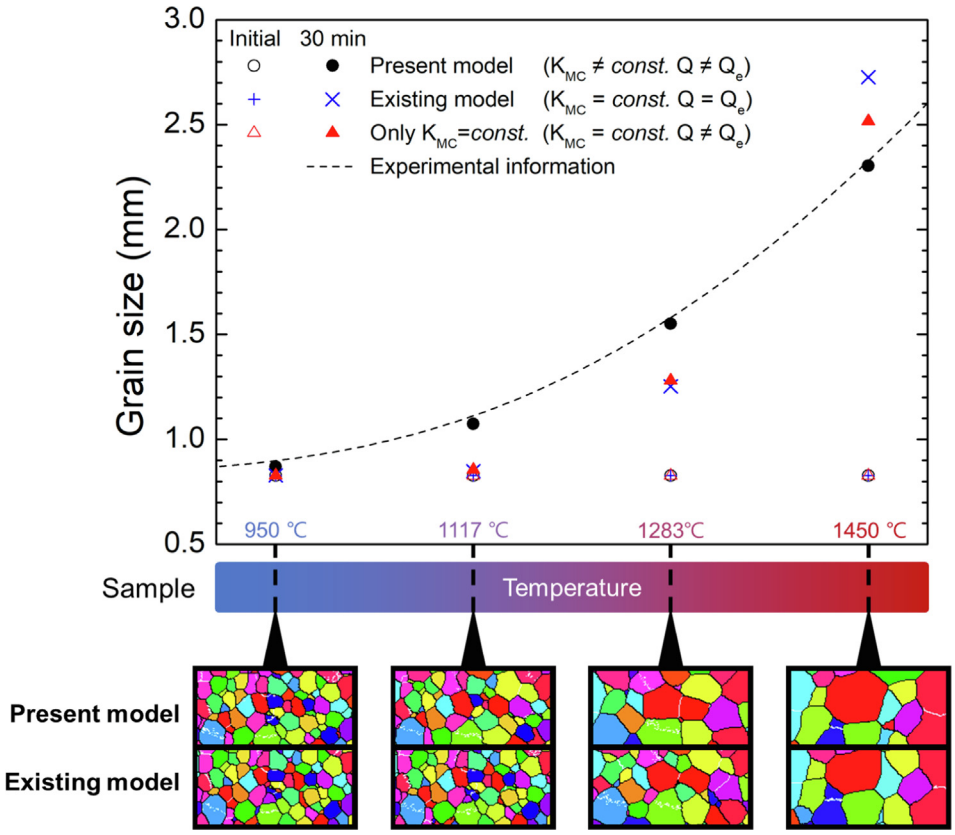



**Fig. 5.** Simulated grain growth for a 304L stainless steel sample at different temperatures. $K_{MC}$ and $Q_e$ value for the "existing model" were obtained from Rodgers *et al.* [28], and "only$K_{MC}$ = *const.*" data were obtained using the same simulation parameters as "present model" and real time conversion scheme of "existing model" (which does not consider the temperature dependence of $K_{MC}$). $K_e^0$ and $Q_e$ values analyzed by Rodgers *et al.* [28] from experimental information [39] were also used to draw the line representing experimental information.

be further utilized to develop new MC Potts models required to simulate AM processes.

## 4. Conclusion

We complemented the real time conversion scheme for the MC Potts simulation, which otherwise results in an illogical time flow under non-uniform temperature conditions. The solution requires appropriate consideration of the temperature dependence in the simulation results ($K_{MC}$: $K_{MC}^0$ and $Q_{MC}$), which is physically meaningful only when the MC Potts model fully considers both the GB mobility and the driving force of grain growth. For this, we modified the MC Potts model to deal with real temperature and realistic energy values in a quantitative manner. Further, we found that the problem in the real time conversion can be completely solved by making the activation energy of grain growth obtained from simulation results ($Q_{MC}$) equal to the experimental value ($Q_e$). This could be achieved by adjusting the activation energy of GB migration ($Q$) in a reasonable range. The modified MC Potts model and the novel scheme for real time conversion were confirmed to be free from the time conversion error and improve the accuracy of quantitative prediction. The MC Potts simulation can now reasonably predict grain growth in various materials under various temperature conditions in a quantitative manner, making it suitable for the design of materials and practical processes (e.g., metal AM) without any time conversion problem.

**Data availability**

The datasets analyzed during the current study will be made available from the corresponding author on request.

## CRediT authorship contribution statement

**Sang-Ho Oh:** Conceptualization, Data curation, Formal analysis, Methodology, Investigation, Validation, Writing – original draft, Writing – review & editing. **Matae Lee:** Data curation, Software, Investigation, Validation, Visualization. **Byeong-Joo Lee:** Conceptualization, Funding acquisition, Methodology, Software, Supervision, Writing – review & editing.

## Data availability

Data will be made available on request.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matdes.2022.111439.

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